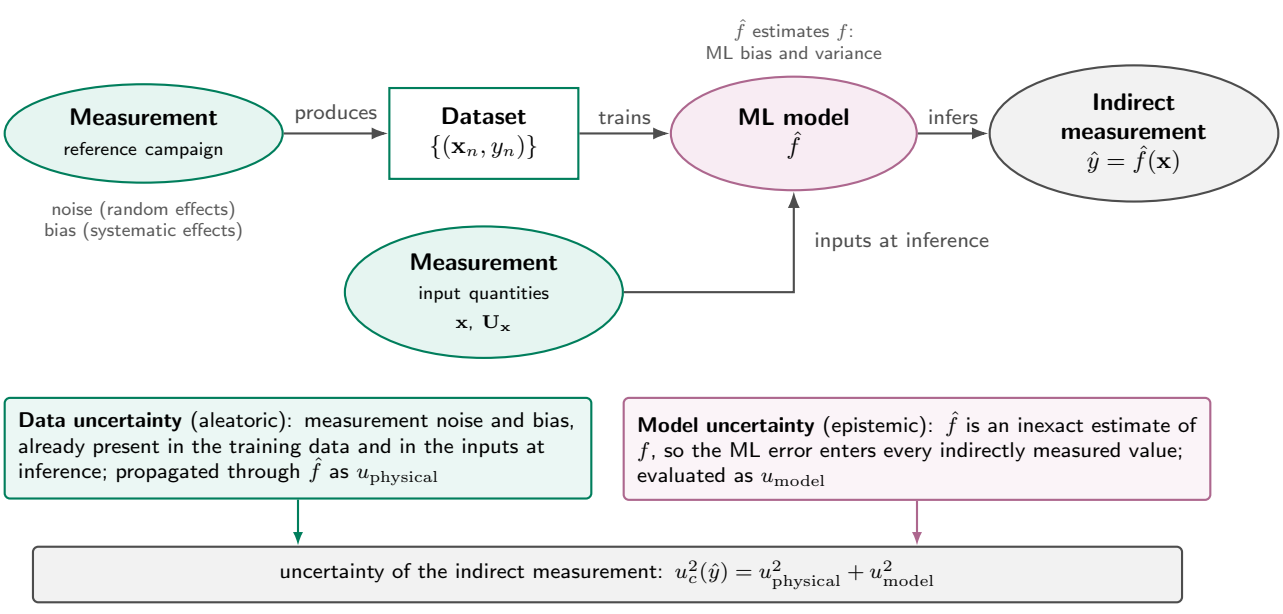

$\hat{f}$ estimates f:
ML bias and variance
Measurement
reference campaign
produces
Dataset
$\{(\mathbf{x}_n, y_n)\}$
trains
ML model
$\hat{f}$
infers
Indirect measurement
$\hat{y} = \hat{f}(\mathbf{x})$
noise (random effects)
bias (systematic effects)
Measurement
input quantities
$\mathbf{x}$, $\mathbf{U}_\mathbf{x}$
inputs at inference
Data uncertainty (aleatoric): measurement noise and bias, already present in the training data and in the inputs at inference; propagated through $\hat{f}$ as u_{physical}
Model uncertainty (epistemic): $\hat{f}$ is an inexact estimate of f, so the ML error enters every indirectly measured value; evaluated as u_{model}
uncertainty of the indirect measurement: $u_c^2(\hat{y}) = u_{\text{physical}}^2 + u_{\text{model}}^2$